

Turion Germination in *Spirodela polyrhiza* Under Varying Temperature Conditions

Abstract

In the experiment of turion germination, the goal was to identify how varying temperatures would impact the turion percent coverage and frond growth from the model organism *Spirodela polyrhiza*. The prediction was that higher temperature treatments would decrease the growth of turions, which established temperature as the independent variable and turion germination as the dependent variable. At 7, 14, and 28 days of the experiment, the fronds were analyzed for percent coverage and frond growth in the Image-J application. These data values were then imported into a coding program, which organized the data into graphs, as well as ran a two-way ANOVA test to determine significance. An overall trend was found that the 20-degree treatment experienced the most successful growth, while the 40-degree treatment experienced the least successful growth. However, no significance was found between the 3 treatments and percent coverage nor the 3 treatments and frond growth. Although no significance was found, this experiment yielded important information regarding the impacts of climate change and provided insight for future research in how to improve the experiment.

Introduction

Climate change is an ecological phenomenon in which temperature averages drastically alter from historical trends (Grimm et al., 2013). Generally, climate change presents the conditions of colder winters and warmer summers. A rise of carbon emissions over the last century has contributed to the relevance of the issue, and the discussion surrounding its impact on ecosystems. Many researchers have found data that shows climate change has already created an impact on present ecosystems, such as by lowering population numbers and disrupting the established ecological flow (Grimm et al., 2013).

Thus, to study the effects of how temperature specifically will affect ecosystems and plant formation, the organism *Spirodela polyrhiza* will be used. *Spirodela polyrhiza* is a species of duckweed that is commonly used as a model organism in research due to its quick reproduction rates, resiliency, and its ability to grow under lab conditions (Appenroth, 2002). In *Spirodela polyrhiza*, the turions are released to the bottom of their water source in unfavorable weather conditions and rise again to the surface once conditions improve (Adamec, 2018). This method of survival allows for the re-population of duckweed following the winter months. However, if temperature in an ecosystem becomes abnormal, this process of re-populating the duckweed may be disrupted, and can be the source of total ecosystem failure, as has been documented by other research of vital organisms experiencing population loss (Grimm et al., 2013).

The goal of this experiment is to study how temperature changes impact the growth of turions in *Spirodela polyrhiza*. Because turions play a crucial environmental role in maintaining the populations of duckweed, the ability of their survival will have implications for other species within the ecosystem. Where other researchers have attempted to answer how several factors affect turion growth, such as phosphate concentration coupled with temperature changes, this experiment focuses solely on temperature as the variable to investigate how varying temperatures as a result of climate change will impact the duckweed (Appenroth, 2002).

In the investigation of how different temperatures (20 degrees, 30 degrees, and 40 degrees) will affect the growth of the turions in the species *Spirodela polyrhiza*, the alternative hypothesis is if exposed to higher temperatures, there will be a significant difference in the growth of turions. The null hypothesis states that if exposed to higher temperatures, there will be no significant difference in the growth of turions. A study done by K.J. Appenroth discerned those higher temperatures usually resulted in a lower percentage of turion growth than occurred in the lower temperatures (Appenroth, 2002). Thus, the prediction for this experiment is that higher temperatures will decrease the growth of turion fronds in the species *Spirodela polyrhiza*.

Methods

Experimental Design

To test the effect of temperature changes on turion growth, three treatments were created, each with 6 replicates. The control group, 20 degrees, was determined from previous research by Appenroth as being a successful temperature for duckweed turions to germinate (Appenroth, 2002). The two experimental groups, 30 degrees and 40 degrees, were chosen to simulate climate change with a goal to either indicate a difficulty in the species' ability to reproduce, or to demonstrate inability to reproduce and/or result in death. A sterile procedure was put into place to create the three microplates. For each plate, 10 mL of sterile plant medium was pipetted into the six wells. After adding the medium, a singular turion was added into each well. The microplates were stored at their respective temperatures.

Data Collection

Once the plates were set up, pictures were taken of each plate to monitor the weekly growth of the turions. The ImageJ program was utilized to count the number of fronds and measure the area of the fronds in each well. Before recording data, the photo was set to scale of 34.8 mm to the number of pixels accounting for the diameter of the well, at an angle of 0 degrees. After being set to scale, the fronds were counted using the multipoint selection and the area of the fronds were calculated using the freehand selection. These values were recorded in an excel sheet.

Data Analysis

To analyze the data, a coding program was used to investigate results of the percent coverage and frond growth data for 3 treatments, Treatment 20, Treatment 30, and Treatment 40, over 28 days. The program created a line graph and boxplot to depict the variables over the 28 days, with means recorded at the 7-, 14-, and 28-day marks, $\pm$ standard error. Two-way ANOVA tests were run to analyze potential significant differences in the data. A p-value of less than 0.05 indicated significance among treatments.

Results

Percent coverage of turion fronds was analyzed under 20 degrees, 30 degrees, and 40 degrees conditions to compare the ability of the turions to germinate in varying temperatures. In both treatments 20 and 30 degrees, a general trend existed that the percent coverage of the fronds increased over the 28 days, while treatment 40 had no increase in percent coverage after day 7 (Figure 1). Treatment 30 also seemingly had the largest increase in percent coverage, although with a large standard of error measured (Figure 1). A two-way ANOVA test was run to analyze the independent variables of treatment and day against each other, which found no significance among the treatments ($p=2.524$, Table 1). Tukey test results were not included due to no significant differences established in data.

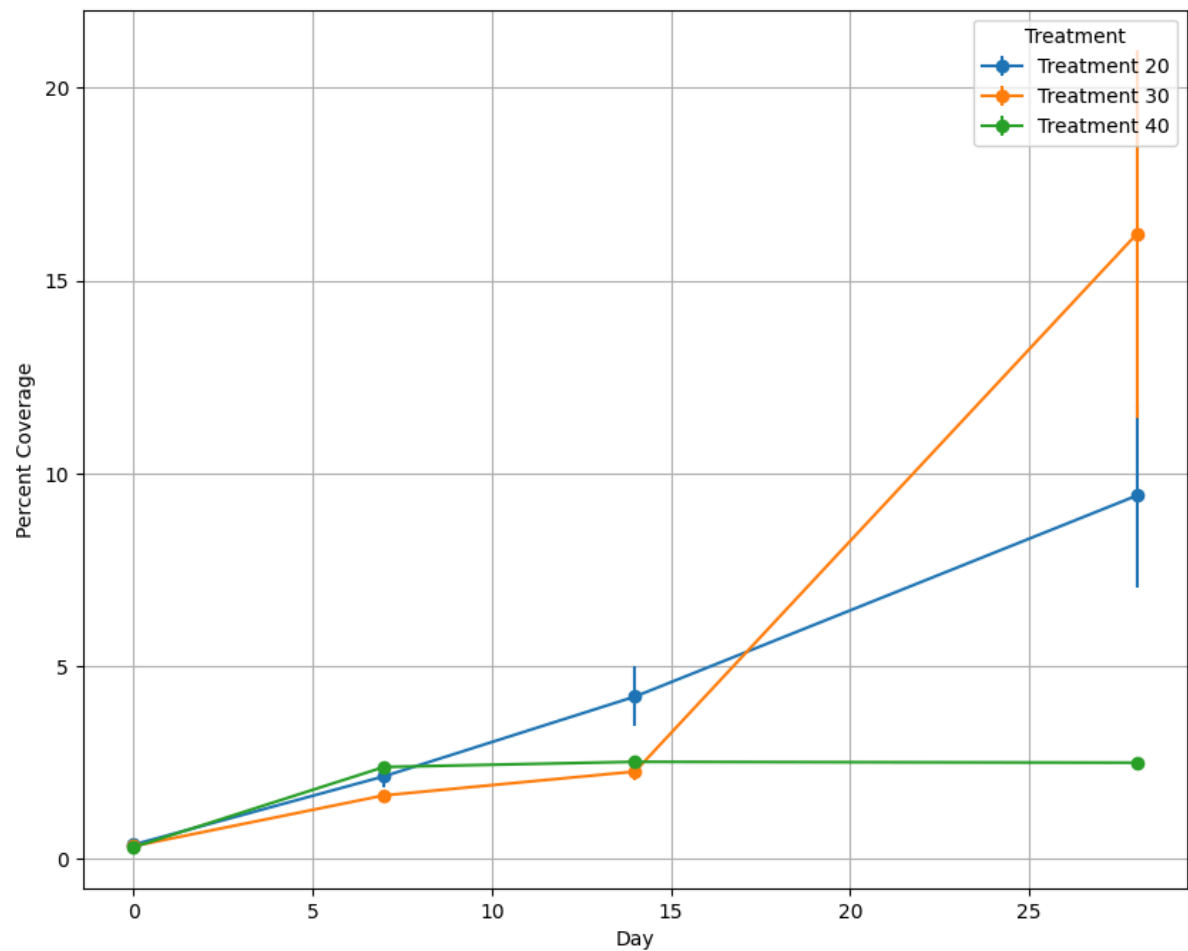


Figure 1. Percent coverage of turions across 28 days for treatments 20 degrees, 30 degrees, and 40 degrees. Six replicates were created for each treatment, with mean values recorded at 7, 14, and 28 days, +/- standard error.

Table 1. Two-way ANOVA data for percent coverage from treatments 20, 30, and 40, each with 6 replicates, measured against day. The p-value is listed in the PR (>F) column, with a p-value < 0.05 being significantly different.

	sum_sq	df	F	PR(>F)
C(Treatment)	126.253005	2.0	4.304628	1.790837e-02
C(Day)	841.133415	3.0	19.119105	8.078122e-09
C(Treatment):C(Day)	452.911050	6.0	5.147372	2.523881e-04
Residual	879.887847	60.0	NaN	NaN

FronD growth was also analyzed under 20 degrees, 30 degrees, and 40 degrees conditions to compare the ability of turions to germinate under varying temperatures. The general trend demonstrated that treatment 20 on average had the largest frond growth out of the 3 treatments, where in comparison treatments 30 and 40 had little overall growth (Figure 2). Treatment 20 also had the largest standard error, while treatments 30 and 40 have smaller standard error; although, treatment 30 presented several outlier points (Figure 2). A two-way ANOVA test was run to analyze the independent variables of treatment and day against each other, which found no significant difference among the treatments ($p=6.413$, Table 2). Tukey test results were not included due to no significant differences established in data.

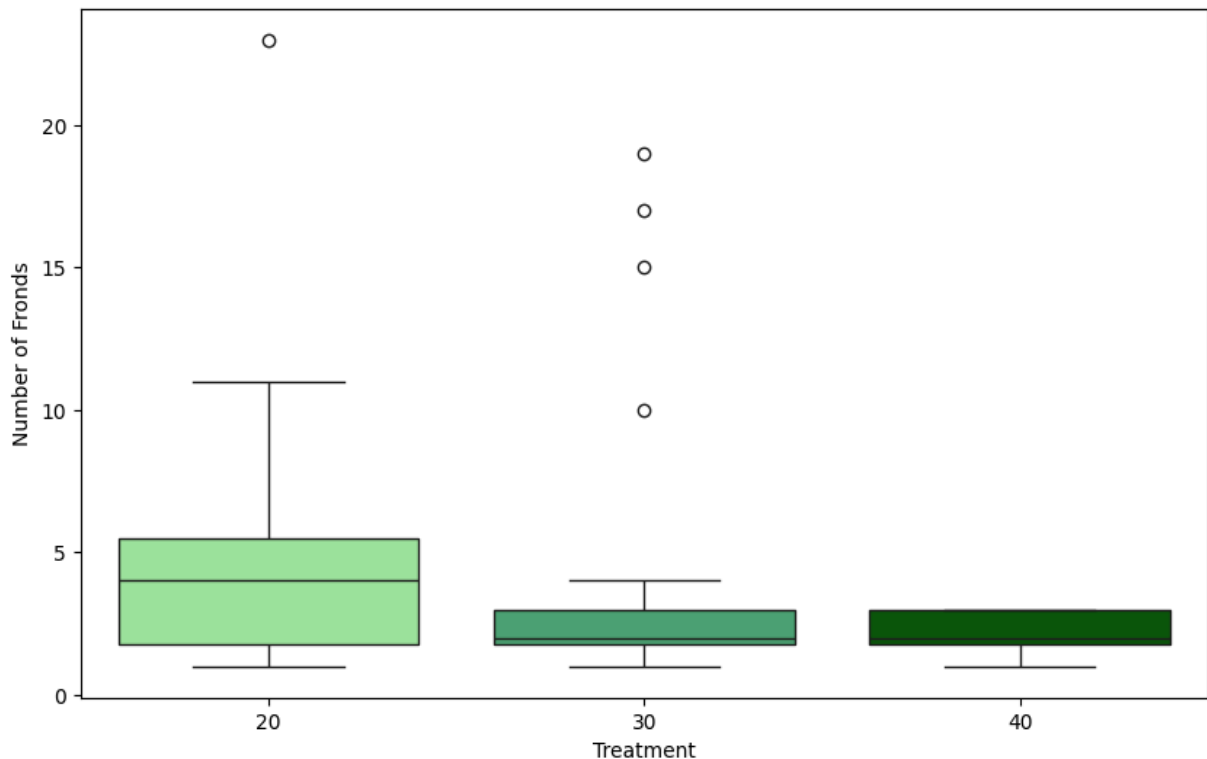


Figure 2. Boxplot representing number of fronds grown across 28 days for treatments of 20 degrees, 30 degrees, and 40 degrees. Six replicates were created for each treatment, with the box representing the middle 50 percent of the data, +/- standard error.

Table 2. ANOVA data for number of fronds from treatments 20, 30, and 40, each with 6 replicates. The p-value is listed in the PR (>F) column, with a p-value < 0.05 being significantly different.

	sum_sq	df	F	PR(>F)
C(Treatment)	110.083333	2.0	7.377141	1.366015e-03
C(Day)	583.444444	3.0	26.066021	6.419344e-11
C(Treatment):C(Day)	206.805556	6.0	4.619633	6.413410e-04
Residual	447.666667	60.0	NaN	NaN

Discussion

This study aimed to determine the effects of varying temperatures on turion frond growth by analyzing the percent coverage and frond number over 28 days. As seen from Figure 1, the general trend presented is that treatments 20 and 30 experienced a larger growth on average than did treatment 40, which no longer experienced growth after day 7. Additionally, Figure 2 displayed the trend that treatment 20 had a larger frond growth when observing the middle 50 percent of data than did treatments 30 and 40. However, as evident in Tables 1 and 2, no significant difference was established between treatments because the p-values were greater than 0.05 and fails to support the prediction, which stated higher temperatures will decrease the growth of turion fronds in the species *Spirodela polyrhiza*. Thus, the alternative hypothesis which claimed a significant difference exists between treatments exposed to different temperatures is rejected; the null hypothesis which claimed that significant difference does not exist between treatments exposed to different temperatures is failed to be rejected as there is not enough data to determine a significant difference.

In a study done by Appenroth, it was found that temperatures in the 20-degree Celsius range yielded higher percent growth of turion fronds than did treatments put under higher temperature conditions (Appenroth et al., 1996). Although the data from the results fails to support a significant difference, the trends observed in the analysis represent a similar finding. Figure 1 demonstrated that the highest frond number growth was experienced in the 20-degree treatment, suggesting this is an ideal temperature for turion growth, while the least frond number growth was experienced in the 40-degree treatment, suggesting that higher temperatures inhibit or kill the turions, and is unfavorable in for their survival. Additionally in Figure 1, the 20-degree treatment saw larger frond number growth than in the 30-degree treatment, suggesting that 20 degrees was more ideal than 30 degrees for turion germination success.

The unexpected results were due to several limitations within this study. As for human error, each treatment was measured by a different person, which may have led to inconsistencies in the usage of the Image-J program as well as how measurements were recorded, leading to discrepancies between treatments. Additionally, the building in use for storing the treatments experienced fluctuating room temperatures, which may have impacted our results, especially of the 20-degree treatment. Should this experiment be replicated, a system should be enforced in

which tasks for each treatment be completed by one person, such as one person measuring the fronds, one person counting the fronds, and one person setting up the procedure within methods. It will also be important for future replications to ensure that the temperature of the storage for each treatment to be accurate without fluctuations.

Although significance was not found within this experiment, the data highlighted how this experiment can be improved in the future to yield better results. The big picture of the study was to produce a finding within the context of climate change. For example, Figure 1 demonstrates how in treatment 40, the fronds died, and growth stopped after about 7 days. This data could be used to study the impacts of how climate change and sequential effects on flora will impact an ecosystem's survival. When replicated, the results could provide credibility to the need for lowering global temperatures, as shown through research by Appenroth, small changes in temperature drastically altered the plants' abilities to survive (Appenroth, 2002). To expand this experiment, fronds should be observed in different types of water, not just sterile water solution. As is demonstrated in the research done by Bökük et al., different minerals within varying water sources may produce a confounding variable and create different results when analyzing turion resistance to heat (Bökük et al., 2013). Thus, a future experiment should investigate how plants will adapt in different types of ecosystems, not just one generalized system.

References

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